

Combining and Forward-Carrying Overlapping Wellbore Surveys: An Open Implementation Demonstrated on Public Volve Data

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Version 2.1 — **2026-09-01** · concept DOI: 10.5281/zenodo.22080773 (v2.0 added the covariance EOU (Bulychenkov 2026, *Covariance EOU*, concept 10.5281/zenodo.22148755, v1 10.5281/zenodo.22148756) and correlated single-station fusion, §3. v2.1 adds an overlap-innovation QC gate before conditioning (§3), the smooth-constraint application for gradient-based planning (§7), and the open-benchmark framing (§8), following peer review by K. Bulychenkov; and points to the dedicated companion paper on the exact interior covariance (DOI 10.5281/zenodo.22232546), stating the interior covariance used here as the exact arc-faithful form.)

Abstract

When two independent surveys cover an overlapping interval of the same wellbore — most commonly a measurement-while-drilling (MWD) survey and a gyroscopic survey — the industry routinely keeps only one and discards the other. The statistical value of combining them has been recognised for over two decades (Ekseth, 1998; Bang, 2019) yet remains under-adopted, and open, reproducible implementations are scarce. This paper does two things with an open-source implementation (welleng), demonstrated end-to-end on the public Volve dataset. First, it *combines* overlapping surveys with the best linear unbiased estimator (BLUE), evaluated on the analytical arc-faithful interior covariance so the result is continuous in measured depth rather than tied to survey stations. Second, it *carries the calibration forward*: where a gyro overlaps the upper part of an MWD run, the gyro constrains the MWD’s systematic (magnetic) error — including terms a magnetic survey cannot observe from its own data — and, because that systematic realisation persists down-hole, the constraint reduces the MWD covariance in the deeper, gyro-free section. On Volve well F-4 the combination reduces the vertical 1-sigma by a factor of 1.4 over the overlap; carrying the gyro-observed declination forward reduces the deep MWD-only lateral 1-sigma unconditionally, and by roughly a factor of two when the same MWD tool is reused below. Third, it generalises the carry to the **covariance EOU** (also referred to as the covariated ellipse of uncertainty in earlier communications; Bulychenkov, 2026): the whole well as one block covariance with off-diagonal cross-station correlation, so conditioning on a single gyro station propagates through the correlation to tighten the entire run — on F-4, about a third at total depth from one station, about a half from the whole overlap — with the forward-carry recovered as its forward-only slice. None of the methods is novel — they adopt established estimation theory, and the covariance-EOU approach is due to Bulychenkov — so the contribution is the open, validated implementation, the analytical/continuous formulation, and the quantified demonstration on public field data.

1. Introduction

The ISCWSA/OWSG error model (Williamson, 2000; ISCWSA Rev 5) propagates a survey tool’s error sources into a per-station position covariance. It treats a single survey; it says nothing about what to do when two surveys of the same hole overlap. In practice a definitive survey is assembled by *selecting* the best tool over each interval and concatenating (a tie-chain), discarding the redundant overlap data.

That the redundant data carries recoverable accuracy is long established. Ekseth (1998) called for “a statistical adjustment theory when redundant surveys are available”; Bang (2019, SPE-195621) sets out the rigorous weighted-averaging method for combining overlapping surveys and a combined-IPM construction with inclination/azimuth-dependent weights (building on earlier weighted-averaging work it reviews); and the ISCWSA record notes the method still “prevents wider adoption” for want of practical, trusted tooling

(60th ISCWSA meeting, 2016). Separately, using an independent gyro to constrain an MWD’s systematic error via statistical estimation was presented at the 49th ISCWSA meeting (2019), where a magnetic declination error of about 0.5 degrees was recovered by comparing an MWD azimuth against a gyro.

The estimation itself is textbook — the same inverse-covariance fusion and Kalman/Gaussian conditioning used routinely in multi-sensor data fusion, geodetic least-squares network adjustment, meteorological data assimilation and aided inertial navigation. What has limited its adoption in wellbore surveying is therefore not the mathematics but three practical barriers: the **correlation bookkeeping** it requires (identifying which error sources the two surveys share, so the shared part is not wrongly reduced — the recurring “correlation is the hard part” caution in the ISCWSA record), the **select-and-concatenate convention** and a well-founded caution about *shrinking* an ellipse of uncertainty, and the **absence of open tooling**. This paper addresses all three.

It is worth stressing that the correlation bookkeeping is not new machinery. It is the *same* shared-error accounting already performed when survey legs are **tied**: joining two sequential runs requires tracking which error sources (the geomagnetic/declination and reusable-sensor systematics) are common across the tie so the downstream leg inherits the correlated uncertainty. Standard survey software — including COMPASS, whose tie handling separates global, systematic and well-by-well terms — already carries this decomposition. What combination and forward-carry add is not a new information requirement but a *different use* of the same shared-term structure: to fuse an overlap (the cross-covariance C of Section 2) rather than to concatenate. The gap is therefore one of convention and tooling, not of theory or of missing information — a survey system able to tie legs already holds what is needed to combine them.

What is missing is not the theory but an *open, reproducible* implementation and a demonstration of the achievable benefit on public data. This paper supplies both, for two operations:

- **Combination** (Section 2): a continuous BLUE fusion of two overlapping surveys.
- **Forward-carry** (Section 3): conditioning a survey’s systematic error on an independent overlapping reference, then carrying the reduced systematic into a reference-free section — the aided-inertial-navigation “calibrate then coast” pattern applied across survey tools.

Both are demonstrated on Volve well F-4 (Section 4) and shipped in welleng (Section 5).

2. Combining overlapping surveys

Two surveys A and B provide independent position estimates over a common interval, with per-station covariances Σ_A and Σ_B . The combined best linear unbiased estimate has covariance

$$\Sigma_c = \Sigma_A - (\Sigma_A - C) (\Sigma_A + \Sigma_B - C - C^T)^{-1} (\Sigma_A - C)^T, \quad (1)$$

where $C = \text{cov}(x_A, x_B)$ is the cross-covariance from the error sources the two surveys *share*. Equation (1) is the standard best linear unbiased estimator for two correlated estimates of one quantity (the cross-covariance / “tracklet” fusion form, Bar-Shalom et al., 2001, §8.4). Error components that are fully common-mode do not reduce; only the independent part is averaged down. When the surveys are independent ($C = 0$ — the case flagged for every Volve survey tool, whose **correlate** attribute is uniformly “N”), it collapses to the classic inverse-variance information add (the weighted-averaging method of Bang, 2019):

$$\Sigma_c = (\Sigma_A^{-1} + \Sigma_B^{-1})^{-1}. \quad (2)$$

It is BLUE, not a Kalman filter (there is no time or sequence — just two estimates of one position).

The two surveys need not be different tool *types*. Two MWD runs combine the same way — for example when a bottom-hole assembly is pulled part-way through a section and a new one re-surveys the already-drilled interval (typically while tripping back to bottom). Because both are magnetic, their **geomagnetic reference error is shared** (same location, time and model) and is kept common-mode through C — it does not reduce; only the **independent** parts (the two tools’ sensor biases and the two BHAs’ interference and sag) average down. The gain is therefore real but smaller than an MWD+gyro combination, where the gyro — being declination-independent — also collapses the geomagnetic term. The same C machinery expresses both cases. *When* the re-survey is run matters, and carries its own value — see Section 3.

Two surveys rarely share exactly the same station depths, so both must be brought to a common measured depth. Linearly interpolating the assembled covariance between stations is not the covariance of the resolved

arc, and mis-states it near doglegs. `welleng` instead evaluates each survey’s covariance at any measured depth **analytically**: each error source is propagated onto the minimum-curvature arc through the partial-leg Rodrigues rotation — the same Rodrigues form of the minimum-curvature interpolation used in `welleng`’s `MinCurve` (Sawaryn and Thorogood, 2005) — so the covariance is an analytical function of arc position rather than an interpolation of the covariance matrix. The combination is therefore continuous and dogleg-faithful — queryable at any depth in the overlap, on or off a survey station. (The linear-interpolation error — survey-interval and geometry dependent — is quantified in the companion paper on the exact interior covariance, DOI 10.5281/zenodo.22232546.)

The interior covariance evaluated here is the **exact** covariance of the resolved minimum-curvature arc — exact at the survey stations and continuous between them, propagating each error source onto the arc rather than interpolating the assembled covariance matrix. Its derivation, and its validation against an interpolated-position Monte-Carlo oracle to about 0.2%, are the subject of the companion paper cited above; the combination results here use it directly.

The implementation validates against an independent stacked generalized- least-squares derivation to a relative tolerance of 10^{-8} (covariance and position), and against a Monte-Carlo oracle (Figure 3A). As an external check, it reproduces the published result of Bang (2019, SPE-195621), whose “rigorous averaging” is precisely this $C = 0$ fusion: combining two independent surveys of similar accuracy reduces the one-sigma by the equal-accuracy limit $1/\sqrt{2}$ — a 29.3% reduction, matching Bang’s stated “up to 29%”. The larger lateral reductions Bang reports (his combined lateral extent about a third of the magnetic-only value) arise from a *realistic vendor* continuous-gyro model whose lateral accuracy exceeds the MWD’s; the standard IS-CWSA/OWSG gyro model is by design conservative — poorer than the MWD model — so the gain realised in practice depends on the actual gyro’s quality, and the demonstration below, which uses the survey tools’ own exported error models, is correspondingly a conservative floor.

Recovering the fused trajectory as a survey

The combination returns a covariance and a best estimate at every measured depth; conventional survey software expects a minimum-curvature (measured depth, inclination, azimuth) listing. Which is easier to recover depends on the *space* in which the fusion is performed.

The direct route — fuse in survey (DIA) space: measured depth is the shared, measured index of both surveys, so combining them reduces to combining inclination and azimuth per station, inverse-covariance-weighted in survey space. The fused quantity *is* the (md, inc, azi) listing — a valid minimum-curvature survey by construction, needing no reconstruction and physically consistent (the straight-line chord between consecutive fused stations never exceeds the measured-depth interval). `welleng` exposes this as `combine_surveys(..., space="dia")`; it is the natural, residual-free way to produce a fused survey.

The alternative — fuse in position (NEV) space, then reconstruct: if the combination is instead done on the NEV position covariance (the continuous BLUE above), the fused quantity is a position path and a survey must be recovered from it. This too is analytical: with the tie-in tangent fixed, each leg’s end tangent is the start tangent *reflected about the chord*,

$$\hat{t}_{i+1} = 2(\hat{c} \cdot \hat{t}_i)\hat{c} - \hat{t}_i, \quad (3)$$

the minimum-curvature chord-bisector property, marching out in one pass (`welleng`’s `survey_from_positions`). But arbitrary fused positions are not exactly a minimum-curvature arc chain — averaging two surveys’ positions leaves consecutive points marginally farther apart than the interval permits, or closer — so the reconstruction carries a small position residual (centimetre-scale per station, well below the metre-scale position uncertainty) that the direct DIA fusion avoids entirely. Where the fused overlap is interior, the reconstructed segment is pinned to the continuing single-source survey at the seam. This position→survey reconstruction is offered as a concept pending field validation; the DIA-space fusion is the preferred route.

In both cases the fused stations are **calculated, not measured** (the fused position is a derived best estimate), measured depth is held exact, and — because the fused covariance is continuous in measured depth (Section 2) — every station carries its own true ellipse of uncertainty. The result is a self-consistent fused survey: (measured depth, inclination, azimuth, covariance) at every station, drawable as a minimum-curvature well path.

3. Carrying the calibration forward

A magnetic MWD survey cannot observe some of its own systematic errors from its own data — magnetic declination and certain sensor biases are degenerate in a magnetic-only least-squares solution. An independent gyro, being declination-independent (it references Earth’s rotation and gravity, not the geomagnetic field), breaks that degeneracy: over an overlap it observes the MWD’s realised systematic error directly.

Crucially, that systematic realisation is *the same* over the whole tool run — it persists into the deeper section the gyro never reached. So the overlap observation reduces the MWD covariance at depth. With the target’s shared systematic sources giving the cross-covariance $\text{cov}(x_{\text{deep}}, z)$ between a deep station and the overlap observation $z = x_{\text{MWD}} - x_{\text{gyro}}$, the conditioned deep covariance is the Schur complement

$$\Sigma'_{\text{deep}} = \Sigma_{\text{deep}} - \text{cov}(x_{\text{deep}}, z) \text{cov}(z)^{-1} \text{cov}(z, x_{\text{deep}}), \quad (4)$$

with $\text{cov}(z) = HH^T + \Sigma_{\text{gyro}}^{\text{overlap}} + \Sigma_{\text{MWD,random}}^{\text{overlap}}$ and H the target systematic sources’ propagation over the overlap. Equation (4) is the Gaussian conditional (Schur-complement) covariance — a single Kalman update step (Bar-Shalom et al., 2001, §5) — i.e. error-state / aided-inertial- navigation conditioning of a persistent bias state; the joint-covariance Schur form is used because it inverts the well-conditioned $\text{cov}(z)$ rather than the ill-conditioned observation-noise matrix. The concept is that of the 49th ISCWSA meeting (2019); the method is standard aided navigation (GPS-denied “coasting” of a calibrated inertial system).

Only sources that persist carry, and persistence is per class: a source is correlated between the overlap and the deep section — and therefore carries — only if it is the *same realisation* in both:

- **Geomagnetic reference (declination, field strength, dip)** is a property of the field model at that location and time, not of the tool or the bottom-hole assembly (BHA). It persists across *any* bit trip, BHA change, or tool change, and it is precisely the error an independent gyro observes best. This carry is therefore **unconditional**.
- **Sensor bias, scale and misalignment** are tool-borne: they persist only if the **same MWD tool** (sensor package) is reused deeper. A new BHA is fine so long as the tool is the same; a *different* tool re-realises these and they do not carry.
- **BHA-borne terms (axial magnetic interference, sag)** re-realise with a new BHA and do not carry across one.

The implementation selects which classes persist (`persist='global'` for the unconditional declination carry; `persist=('global','systematic')` when the tool is reused); a non-persisting source has independent deep/overlap realisations, contributes nothing to the cross-covariance, and is left at its full value.

Observability is also geometry-dependent: the overlap constrains only the modes it *excites*; a term that matters at depth but is not exercised by the overlap geometry (a different inclination/azimuth regime) is not carried. The benefit is bounded by the overlap’s observability, not its length alone.

Timing the re-survey — retrospective versus prospective: the forward-carry recasts an operational choice. A re-survey run while tripping *out*, after the section is drilled, improves only the *final* definitive — a retrospective gain. A re-survey run while tripping *in* — the overlapping pass an independent tool (gyro, or simply a fresh MWD) makes over the already-open hole *before* drilling ahead — lets the reduced systematic be **banked and carried forward into the interval about to be drilled**. The tighter uncertainty is then available in real time, while it can still change decisions (anti-collision separation, target sizing, an intersect approach), not merely at the end of the well. Where survey accuracy is a live drilling requirement rather than an end-of-well record, the way-in overlap is worth its rig time precisely because Equation (4) makes the gain prospective. This is the same combine-then-carry machinery; only *when* the overlap is acquired changes what it can influence.

The implementation is validated by an independent Monte-Carlo gate in which the systematic **re-realises** between overlap and deep (a different tool), so only the persisting (global) source may carry: simulating the true target and reference errors, forming the overlap observation, and estimating the deep error with the BLUE gain, the empirical residual covariance matches Equation (4) (Figure 3B).

The covariance-EOU generalisation — the solid covariance and single-station fusion

The forward-carry of Section 3 is one direction of a more general object. The conventional ellipse of uncertainty treats a well as a string of *independent* per-station ellipsoids — a block-*diagonal* covariance. But the shared systematic sources (declination, a reused sensor bias) are a *single realisation* spanning many stations, so

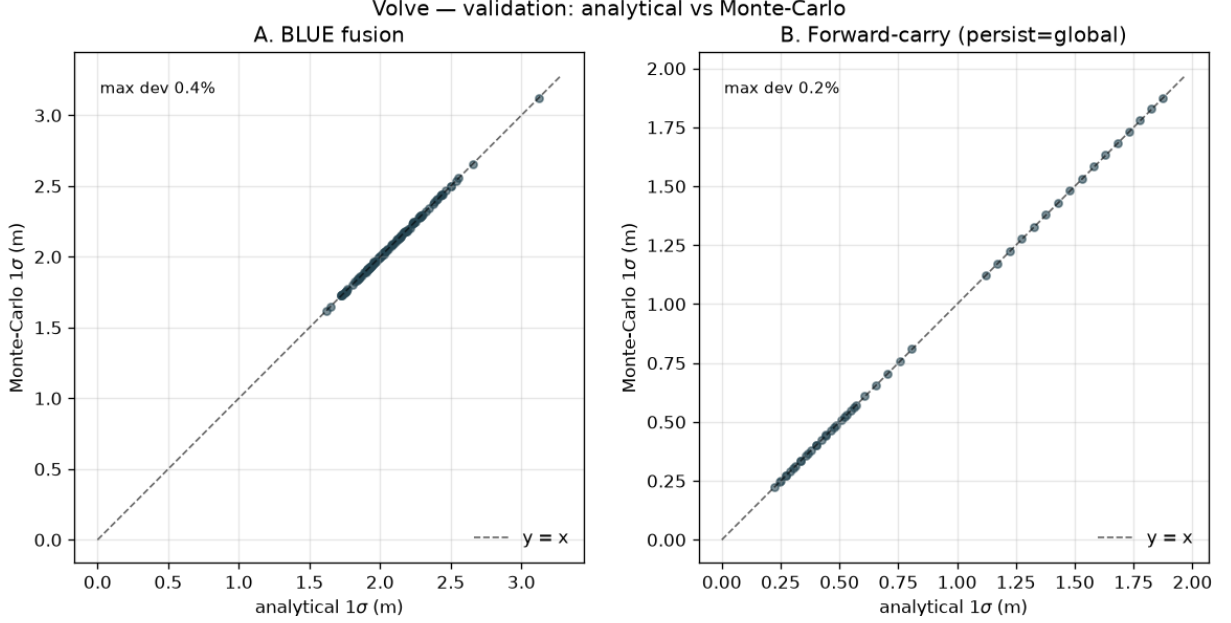


Figure 3. Validation — analytical prediction versus Monte-Carlo empirical 1-sigma, on the line $y = x$. (A) BLUE combination over many random shared/independent covariance cases. (B) forward-carry with the systematic re-realising between overlap and deep (persist = global). Both lie on $y = x$ within Monte-Carlo noise.

the per-station errors are correlated. Retaining that correlation gives the well’s **joint** covariance: one block matrix $\mathbf{C}_{\text{well}}$ whose diagonal blocks are the usual per-station covariances and whose **off-diagonal** blocks are the cross-station correlation. This “solid” covariance is the covariance EOU of Bulychenkov (2026), who casts it as a matrix expansion of the conventional per-station calculation (Williamson, 2000): the weighting function becomes a Jacobian, the error magnitude and propagation mode become a per-source covariance, and the station-wise sum becomes an integration operator:

$$\mathbf{C}_{\text{well}} = \sum_e \mathbf{J}_e \mathbf{C}_e \mathbf{J}_e^T. \quad (5)$$

The off-diagonals are what let information propagate. Conditioning the joint covariance on an independent observation — a gyro station, say — is the standard maximum-likelihood / GLS update (Bar-Shalom et al., 2001),

$$\mathbf{V}_{\text{fused}} = (\mathbf{H}^T \mathbf{C}^{-1} \mathbf{H})^{-1} \mathbf{H}^T \mathbf{C}^{-1} \mathbf{V}. \quad (6)$$

Because the systematic is one realisation across every station it spans, the reduction does not stay at the observed station: it propagates through the off-diagonal blocks to the *whole* correlated well. A single gyro station thus tightens the entire run — most where the systematic has accumulated into a large position error, i.e. deep. The forward-carry of Section 3 is exactly the forward-only slice of this: there the calibration is carried explicitly into the deeper section; here the correlation carries it in every direction it reaches (run, well, and — across wells sharing a geomagnetic reference — pad-wide), with no separate carry step. The declination cross-well correlation follows a partial- correlation table rather than an artificial added term.

The same off-diagonals that let one good station tighten the whole run will let one *bad* station corrupt it: an erroneous observation propagates through the correlation just as a good one does. A quality check on the observation must therefore precede the conditioning, and it belongs in the reference implementation rather than the caller. Before the update is applied, welleng gates the overlap innovation $d = x_B - x_A$ on its own covariance

$$S = \Sigma_A + \Sigma_B - C - C^T. \quad (7)$$

Under the null hypothesis that the two surveys agree within their stated uncertainty, d is zero-mean Gaussian with covariance S , so the Mahalanobis distance $d^T S^{-1} d$ is chi-square with three degrees of freedom. A station

exceeding a chosen quantile (the 99.9th percentile by default) is flagged: the two surveys disagree by more than their combined uncertainty allows — a bad station, or an under-stated input covariance — and the joint conditioning should not be trusted until it is resolved. This is a consistency gate, not an outlier rejection: it surfaces the disagreement for the analyst rather than silently down-weighting it.

welleng implements this as the scalar reference: `covariance_block` assembles $\mathbf{C}_{\text{well}}$ (each correlated source contributing a rank-one cross-station block, each random source a diagonal block; its diagonal reproduces the conventional per-station covariance to machine precision), and `fuse_covariates` performs the conditioning. The conditioning is validated by an independent Monte-Carlo gate: simulating the true correlated errors, forming the gyro observation, and applying the update, the empirical residual covariance matches the analytical posterior to within Monte-Carlo noise. On Volve F-4 (Section 4), a single gyro station in the overlap reduces the total-depth position 1-sigma by about a third; the gyro over the whole overlap reduces it by about a half — the deep MWD-only section inheriting the gyro-constrained systematic through the correlation alone. As with the forward-carry, the gain scales with the gyro’s real accuracy advantage over the correlated (declination-limited) part of the MWD budget; a conservative standard gyro model shows little, a representative vendor model the figures above.

An arc-faithful covariance-EOU surface off-station: combining two surveys requires their covariances on a *common* measured depth, and the two tools rarely share a station grid. Bulychenkov’s integration operator Ψ (2026, Eqs 25–28) already propagates the DIA covariance to points of interest along the well, so off-station evaluation is part of his framework; that operator is the same balanced-tangential accumulation welleng propagates with. welleng’s refinement is in the *diagonal* blocks: it forms them from its analytical **arc-faithful interior** covariance (`cov_nev_at`, Section 2) — the exact partial-leg propagation — instead of a full-leg/linear interior, which is exact mid-leg where a linear interior is not (Section 2). Paired with the correlated sources’ cross-station structure for the off-diagonals (`covariance_block_at`), the result is an off-station joint covariance whose diagonal stays accurate through doglegs. It is also what makes the combination *consistent*: because both surveys are evaluated at the same query depth, the result is order-independent — combining survey A with B equals combining B with A to numerical precision, whereas resampling one survey onto the other’s station grid would not be. The arc-faithful interior covariance is welleng’s own (Section 2); pairing it with the covariance-EOU block’s cross-station structure is the accuracy-relevant contribution of this implementation.

Scope: this paper applies the covariance-EOU structure to the standard ISCWSA/OWSG toolcode error budget (Section 4’s per-tool IPMs). Bulychenkov’s full method goes further, replacing certain toolcode terms with dedicated realistic error models — an MSA-correction covariance (from the linearised multi-station system), a BHA-sag-correction covariance (by Monte-Carlo), and a survey-frequency covariance — which this paper does not apply. The figures here are therefore on the standard toolcode budget and are not directly comparable to results from his full realistic-error model.

The same joint covariance underlies a cross-well separation factor for anti-collision (subtracting the shared covariance between two wells), which is beyond this paper’s scope. The covariance-EOU method and these applications are due to Bulychenkov (2026); the contributions here are an open, reproducible implementation, its validation on public field data, and the arc-faithful off-station covariance-EOU surface built on welleng’s analytical interior covariance.

4. Demonstration on public Volve data (well F-4)

Volve well F-4 carries an MWD survey over the whole run and a stationary gyro (“Wellbore Surveyor”) overlapping the upper part, with MWD continuing alone below about 2640 m. All error models are the exact per-tool IPMs imported from the public Volve EDM export; the geomagnetic reference is read from the export’s `DP_MAGNETIC` record, so magnetic inputs are identical to the operator’s. The whole demonstration is reproduced by `examples/survey_combination_volve_f4.py` (Section 5).

Combination (Figure 1): over the overlap the fused 1-sigma sits below both input surveys on every axis. At the base of the overlap the reduction relative to the better single survey is 1.2 (north), 1.3 (east) and 1.4 (vertical).

Forward-carry (Figure 2): two cases, by what is assumed to persist. The **declination carry alone** — unconditional, requiring no assumption about the deeper BHA or tool — reduces the deep lateral 1-sigma by about 1.2 (north, east) at total depth. If the **same MWD tool is reused** below (a new BHA is fine), the sensor systematic carries too and the deep lateral 1-sigma falls by roughly a factor of two — 2.0 (north), 2.2 (east). The reduction persists through the entire gyro-free section, because it acts on the systematic realisation the gyro constrained, not on any measurement the gyro made at depth. (In F-4 the deep section

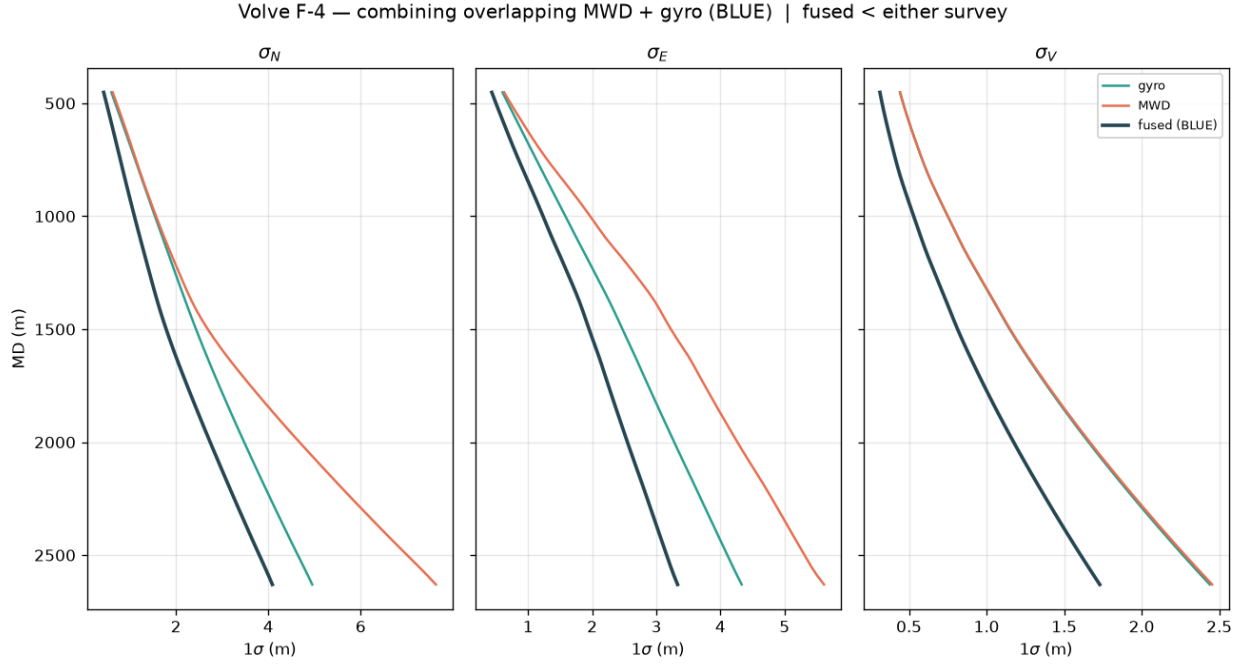


Figure 1. Combining overlapping MWD and gyro on Volve F-4 (BLUE). The fused curve is below either input on all three axes.

crosses the 12-1/4-inch to 8-1/2-inch hole-size change, so the same-tool case assumes the operator re-ran the same MWD sensor package below the shoe — common practice, but an assumption; the declination carry needs no such assumption.)

5. Open implementation

The methods are in welleng (open source):

- `welleng.combination.fuse_covariances` — BLUE fusion (Equation 1), with the cross-covariance for shared terms and a positive-semidefinite guard.
- `welleng.combination.combine_surveys` — continuous combination via the analytical interior covariance `Survey.err.cov_nev_at`. `space="dia"` fuses in survey space and returns the fused (md, inc, azi) listing directly (measured depth held exact, no reconstruction); `space="nev"` (default) fuses positions, with `return_trajectory=True` recovering a survey via the reflection inverse.
- `welleng.utils.survey_from_positions` — the analytical (reflection) inverse of the minimum-curvature method: (md, inc, azi) from a fused position path and a fixed tie-in tangent (the NEV-space alternative).
- `welleng.combination.carry_systematic_forward` — the Schur conditioning of Equation (4) (the forward-only slice of the covariance-EOU conditioning).
- `welleng.combination.covariance_block` / `covariance_block_at` — the covariance-EOU (“solid”) NEV covariance at survey stations / at arbitrary MDs (the continuous surface: arc-faithful interior diagonals + cross-station off-diagonals).
- `welleng.combination.fuse_covariates` — conditioning the joint covariance on a gyro observation (Section 3): the single-station-lifts-the-whole-well update. Implements the covariance-EOU method of Bulychenkov (2026).

Each is a scalar reference implementation (one survey pair in, one result out), with the correctness gates described above kept as regression tests.

The example `examples/survey_combination_volve_f4.py` reproduces every number and figure in Section 4 (run it, adding `--plot` for the figures). It uses only public data: the F-4 survey, the two exact per-tool IPM error models and the magnetic reference are packaged with the repository (derived from the public Volve EDM export, CC BY 4.0), so no external download is required.

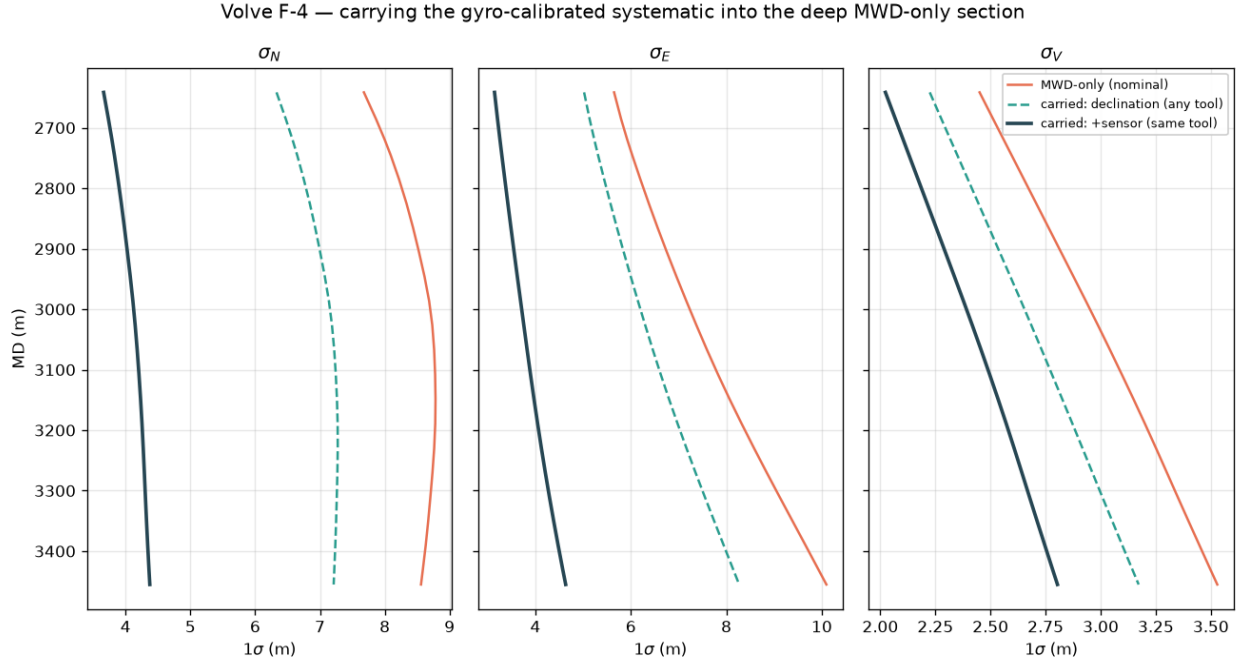


Figure 2. Carrying the gyro calibration into the deep MWD-only section on Volve F-4. Dashed: declination only (unconditional). Solid: plus sensor systematic (same MWD tool reused). Both stay below the nominal MWD-only covariance throughout.

6. Limitations

The combination assumes the shared/independent decomposition between the two tools is modelled correctly; the Volve demonstration uses the export’s own `correlate = N` (fully independent), which is optimistic where the two tools in fact share a tie-on or grid-convergence term. The forward-carry benefit is observability-bounded and persistence-bounded (Section 3): only the geomagnetic reference carries unconditionally, while the sensor systematic carries only if the same MWD tool is reused below — an operational assumption the analyst must confirm, and one the EDM export does not record (it carries no tool serial). Both are demonstrated on one public dataset; broader field validation is future work.

7. Suggestions and future directions

The following are **suggestions**, not validated results — offered because they follow naturally from the mechanism, but each needs work before it could be claimed.

- **Shared-surface-location declination (an IFR-from-gyro suggestion).** Wells sharing a surface location — a platform, a subsea template, or a land pad — share essentially the same geomagnetic model error, so a gyro run on the first well constrains a declination error common to the others, and the correction could in principle benefit them too. This is a form of **in-field referencing (IFR)** — established practice, usually from a magnetometer field survey — derived instead from a gyro’s true azimuth. The comparison must be precise (Jamieson, ISCWSA eBook): standard IFR’s deliverable is the **crustal correction**, which “vary only on geological timescales and therefore can be considered fixed over the lifetime of the field” — so it is **not** time-dependent, and neither is a gyro-derived equivalent over that field. The main-field secular variation is tracked separately by the geomagnetic reference model (BGGM in commercial workflows; welleng’s own client uses the free WMM/IGRF, and for this reproduction reads the operator’s stored reference from the EDM directly). The genuinely time-variant part is the **diurnal / ring-current disturbance**, corrected in real time by **IIFR** (interpolated IFR). A gyro therefore captures the total azimuth error (crustal + main-field + the diurnal *at its run time*) directly, and could substitute for the **crustal IFR** correction over a platform while also delivering an independent survey and the forward-carry — but it does **not** replace **IIFR’s real-time diurnal correction** for wells drilled at other times, nor IFR’s **spatially-mapped** crustal correction over a large field. Realising the platform-scale idea would require sharing the geomagnetic- reference source across wells’ error models; it should be framed against existing IFR/IIFR methods, not as a new concept.

- **Cost (an open question).** Where a gyro is run for its own sake, the declination information is effectively a byproduct; leveraging it across contemporaneous platform wells could be a cost-effective complement to a dedicated IFR service. Whether it is cheaper overall is operator- and case-specific (a standalone gyro run is rig time versus an IFR data service) and is not assessed here.
- **Correlation modeling.** The Volve demonstration uses the export’s `correlate = N`; a per-error-source shared/independent map between the two tools (tie-on, grid convergence) would refine the combination where it is not.
- **Smooth separation-factor constraints for automated planning.** Beyond accuracy near doglegs, a covariance that is continuous in measured depth makes a separation-factor constraint smoother along the trajectory. Station-wise linear interpolation of the covariance makes the constraint non-smooth exactly at the survey stations, and those kinks degrade the convergence of the quasi-Newton and SQP solvers used in gradient-based well planning. The arc-faithful covariance removes them: minimum curvature is C^1 and curvature still jumps at stations, so the claim is smoothness of the constraint *gradient*, not more — but that is enough to help an optimiser. Whether this materially improves planner behaviour is untested here and offered as a direction (the observation is due to K. Bulychenkov, in review).

8. Conclusion

Combining overlapping surveys, and carrying an independent survey’s calibration forward into a reference-free section, recover substantial wellbore-position accuracy that current select-and-discard practice leaves on the table — a factor of about two on the deep lateral uncertainty in the Volve F-4 example. The estimation is textbook; the contribution here is an open, reproducible implementation, an analytical continuous formulation, and a quantified demonstration on public field data.

Beyond the specific result, an open and validated implementation is worth stating as an end in itself. ISCWSA publishes static test wells for verifying error-model implementations; an open, reproducible implementation of survey combination, fusion and — with the covariance-EOU separation factor — anti-collision is their executable counterpart: a reference against which commercial engines can be cross-validated, rather than each being checked only against itself. That is the standard we intend welleng to hold to here.

Acknowledgements

The covariance EOU is the method of **Konstantin Bulychenkov** (MWD STD), set out in his *Covariance EOU* framework (2026) and cited throughout Section 3 — the joint covariance, the integration operator, the MLE MWD-gyro fusion, the OSF, and the realistic-error models are all his. We thank him for generously sharing and discussing the work, and for a careful peer review of this paper — the overlap-innovation QC gate (§3), the smooth-constraint application (§7) and the open-benchmark framing (§8) were all prompted by his review. The contribution of this paper is limited to an open, reproducible implementation of the covariance-EOU approach, the arc-faithful off-station surface built on welleng’s analytical interior covariance, and validation on public field data. Volve data © Equinor and the Volve licence partners, released under CC BY 4.0 (attribution: Equinor and the Volve field licence partners).

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